Gravitation

Law of Gravitation

Gravitational Force :- Gravitational forces are one of the four fundamental forces in nature. These forces are the weakest forces in nature.

There are certain properties of gravitational forces-

(A) It is a central force.

(B) It obeys the inverse square law.

(C) It is always attractive in nature.

(D) It is a long-range force.

(E) The graviton is the field particle of gravitational force.

(F) It is the weakest force operating in nature.

Universal Law of Gravitation :-

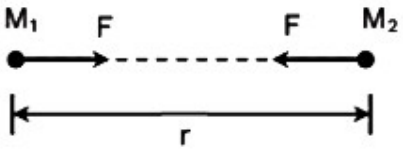
Sir Isaac Newton discovered the universal law of gravitation. As per universal law of gravitation, the force of attraction between two bodies of masses M_1 and M_2 separated by a distance r is found to be –

(A) Directly proportional to the product of masses.

$$F \propto M_1 M_2$$

(B) Inversely proportional to the square of distance between the masses.

$$F \propto \frac{1}{r^2}$$



So, it can be concluded that,

$$F \propto \frac{M_1 M_2}{r^2}$$

We need a constant of proportionality to remove the sign of proportionality.

Newton termed this constant as G (Universal Gravitational Constant).

So, $\boxed{F = \frac{GM_1M_2}{r^2}}$

Value of G is found to be 6.67×10^{-11} Nm^2/kg^2 experimentally. This value is constant throughout the universe.

Dimensional Formula of G:-

$$\text{Unit of } G = \frac{Nm^2}{Kg^2}$$

$$\text{Dimensions of } G = \frac{[MLT^{-2}][L^2]}{[M^2]} = [M^{-1}L^3T^{-2}]$$

Q. **Two bodies having masses 1200 kg and 250 kg are kept at a distance 1 m from each other. What is the gravitational force of attraction between them?**

Sol. Gravitational force, $F = \frac{GM_1M_2}{r^2} = \frac{6.67 \times 10^{-11} \times 1200 \times 250}{1^2}$

$F = 6.67 \times 3 \times 10^5 \times 10^{-11}$

$= 20.01 \times 10^{-6}$ N

Q. **What would happen to the gravitational force between two masses if the separation between them is quadrupled?**

(A) remain unchanged

(B) become one-fourth

(C) become one-sixteenth

(D) become double.

Sol. one-sixteenth, $F \propto \frac{1}{r^2}$

On doubling r, $r' = 2r$

$$F' \propto \frac{1}{r'^2} \propto \frac{1}{(4r)^2} \propto \frac{1}{16r^2}$$

So, $$\frac{F'}{F} = \frac{r^2}{16r^2} = \frac{1}{16}$$

$$\boxed{F' = \frac{F}{16}}$$ $\Rightarrow$ Force becomes one-sixteenth.

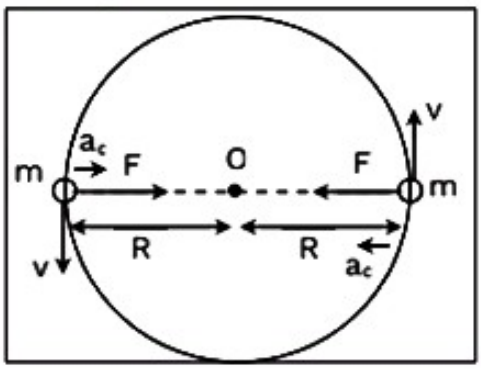
Q. **If two particles having equal masses are moving around a circle of radius R under the effect of their mutual gravitational force, then what is the speed of each particle?**

Sol. The gravitational force acts a centripetal force for the particles.

So, $F = ma_c$

$$\frac{Gmm}{(2R)^2} = m\frac{v^2}{R}$$

$$\frac{Gm}{4R^2} = \frac{v^2}{R} \quad \Rightarrow \quad v = \sqrt{\frac{Gm}{4R}} = \frac{1}{2}\sqrt{\frac{Gm}{R}}$$



Q. **Two spheres made up of same material having radius R each are kept in contact with each other. The force of gravitation between them is proportional to**

Sol. According to Newton's law of gravitation,

$$F = \frac{Gm^2}{d^2} = \frac{G\left(\frac{4}{3}\pi R^3\right)^2}{(2R)^2} = \frac{4}{9}\pi^2 G R^4$$

So, $F \propto R^4$

Principle of Superposition of Gravitational Forces

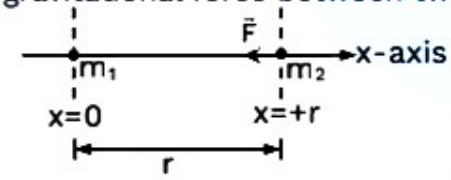
For a collection of point masses, the gravitational force between any two point masses is independent of the presence of other masses.
The net force on any mass is to be added vectorially.
For a system of multiple masses m_1, m_2, m_3 and m_4 , the net force on the mass m_1 due to the other masses in the system is written as,

$$\vec{F}_{net} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14}$$

Q. **Two point masses m_1 and m_2 are taken under consideration. m_1 is kept fixed at the origin and the other mass m_2 is placed at different points. Find the force vector acting on the mass m_2 in each case**
(i) (r, 0) **(ii) (0, r)** **(iii) (r, r)** **(iv) (0cosθ, rsinθ)**

Sol. The gravitational force between two masses is always attractive in nature.

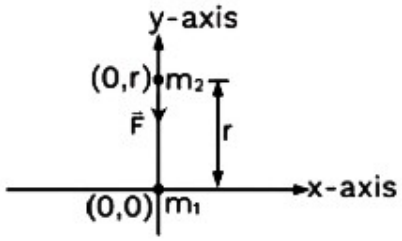
(i)



Gravitational force on m_2 will be towards -ve x-axis.

So, $\vec{F} = \frac{Gm_1m_2}{r^2}\hat{i}$

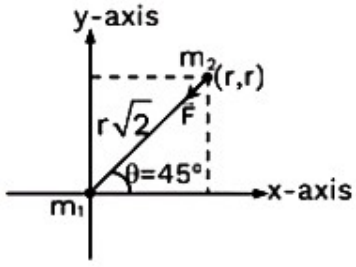
(ii)



Gravitational force on m2 will be towards -ve y-axis.

Clearly, $\vec{F} = \frac{Gm_1m_2}{r^2}\hat{j}$

(iii)



The gravitational force on m_2 will be along the line joining m_1 and m_2 as shown in the above figure.

Separation between m_1 and m_2 = $\sqrt{2}r$

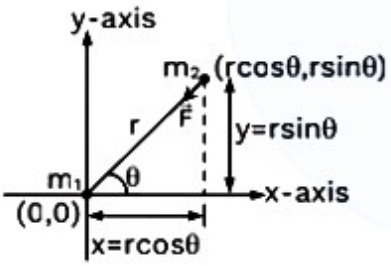
And, the angle $\theta = 45°$.

So, the magnitude of force, $|\vec{F}| = \frac{Gm_1m_2}{(\sqrt{2}r)^2} = \frac{Gm_1m_2}{2r^2}$

In vector form, the force on m_2, $\vec{F} = -|\vec{F}|\cos\theta\hat{i} - |\vec{F}|\sin\theta\hat{j}$

$$\vec{F} = -\frac{Gm_1m_2}{2\sqrt{2}r^2}(\hat{i} + \hat{j})$$

(iv)



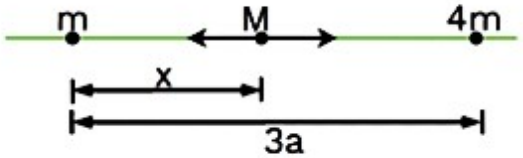
eparation between m_1 and m_2 = $\sqrt{(r\cos\theta)^2 + (r\sin\theta)^2} = r$.

Magnitude of force, $|\vec{F}| = \frac{Gm_1m_2}{r^2}$.

Force in vector form, $\vec{F} = -\frac{Gm_1m_2}{r^2}(\cos\theta\hat{i} + \sin\theta\hat{j})$

Q. **Two point masses 4m and m are kept at a distance 3a apart. At what distance from the mass m a third mass can be kept so that it experiences no net gravitational force on it.**

Sol. For net force on the third mass to be zero, it should be kept in between the two given masses so that their attractive forces balance each other.



As net force is zero, so $\vec{F}_1 + \vec{F}_2 = 0 \Rightarrow \frac{Gmm_0}{x^2} - \frac{4Gmm_0}{(3a-x)^2} = 0$

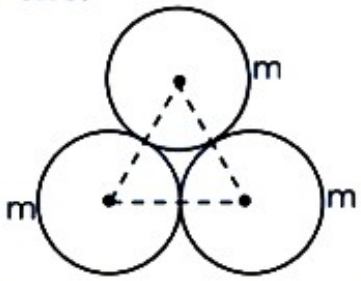
$$\frac{1}{x^2} = \frac{4}{(3a-x)^2}$$

$$3a - x = 2x$$

$$\Rightarrow x = a$$

so, the third object must be kept at a distance "a" from the mass m.

Q. **Three uniform solid spheres (each having mass m and radius R) are kept such that they are touching each other. What is the force of attraction on any one sphere due to other two.**



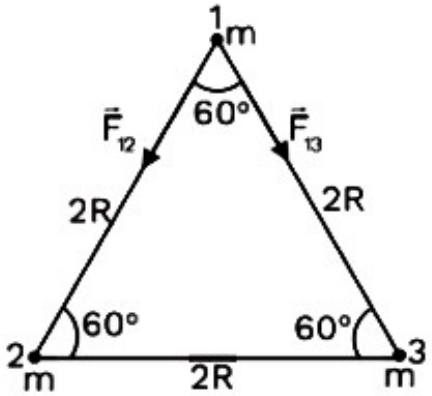
Sol. Due to spherical symmetry, all the spheres can be assumed to be point masses at the corners of an equilateral triangle of side 2R.

$$\left|\vec{F}_{12}\right| = \left|\vec{F}_{13}\right| = \frac{Gmm}{(2R)^2} = \frac{Gm^2}{4R^2}$$

$$\vec{F}_{net} = \vec{F}_{12} + \vec{F}_{13}$$

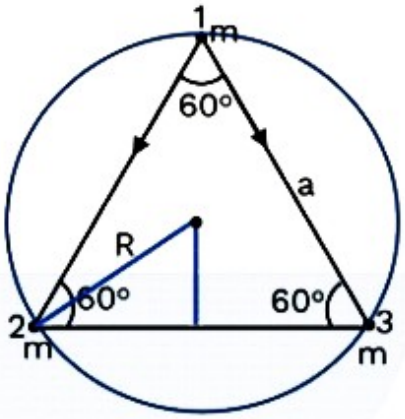
$$\Rightarrow \quad F_{net} = \sqrt{F^2 + F^2 + 2FF\cos 60^\circ} = \sqrt{F^2 + F^2 + F^2} = \sqrt{3}\,F$$

$$= \frac{\sqrt{3}Gm^2}{4R^2}$$



Q. **3 identical particles of mass m are at the vertices of an equilateral triangle of side "a". What is the velocity of each particle if the particles are moving on a circular path without changing the side length of the equilateral triangle?**

Sol.



Net force on any mass along the centre of the circle, $F_{net} = \frac{\sqrt{3}\,Gm^2}{a^2}$

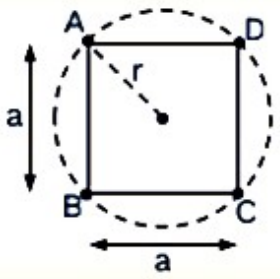
This force acts as centripetal force the circular motion.

So, $\frac{mv^2}{R} = \frac{\sqrt{3}\,Gm^2}{a^2}$

Also, $\cos 30° = \frac{a}{2R} \Rightarrow R = \frac{a}{\sqrt{3}}$

Then, $\frac{\sqrt{3}\,mv^2}{a} = \frac{\sqrt{3}Gm^2}{a^2} \Rightarrow v = \sqrt{\frac{Gm}{a}}$

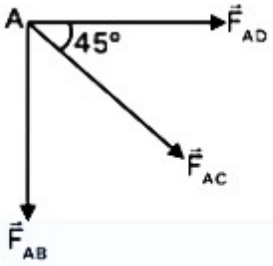
Q. **What is the speed of each particle kept at the corners of a square of side a moving in a circular path under action of their mutual gravitational force?**



Sol. From the figure, $r = \frac{a}{\sqrt{2}}$ (from geometry)

Centripetal force for circular motion = $\frac{mv^2}{r} = \frac{mv^2}{a}\sqrt{2}$ ...(i)

The gravitational attraction on any body due to all other bodies provides necessary centripetal force.



Force on A,

$$\left|\vec{F}_A\right| = \left|\vec{F}_{AB} + \vec{F}_{AC} + \vec{F}_{AD}\right|$$

$$\Rightarrow \quad \left|\vec{F}_A\right| = \frac{Gm^2}{(a\sqrt{2})^2} + \sqrt{2}\frac{Gm^2}{a^2}$$

$$= \frac{Gm^2}{a^2}\left(\frac{1}{2} + \sqrt{2}\right)$$

$$= \frac{Gm^2}{a^2}\left(\frac{1+2\sqrt{2}}{2}\right) \quad \text{...(ii)}$$

From (i) and (ii)

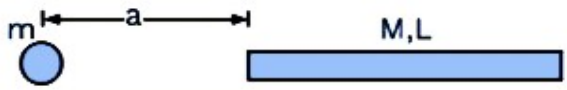
$$\frac{mv^2}{a}\sqrt{2} = \frac{Gm^2}{a^2}\left(\frac{1+2\sqrt{2}}{2}\right)$$

$$\Rightarrow \quad v = \sqrt{\frac{Gm}{a}\left(\frac{1+2\sqrt{2}}{2\sqrt{2}}\right)}$$

Continuous Mass Distribution

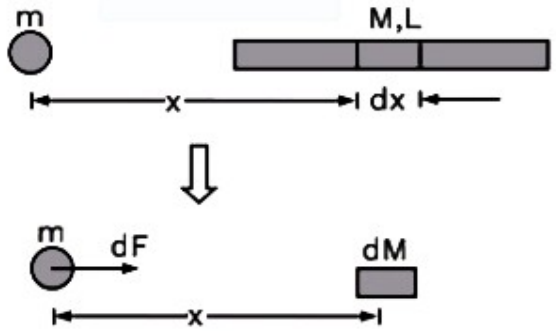
The Newton's law of gravitation is directly applicable for point masses only.
In order to find the gravitational force acting between continuous bodies, we need to divide them into infinitesimal point masses and then use integration to find the net force.

Q. **Find the gravitational force between a point mass m and a rod of mass M and length L.**



Sol. As the rod is not a point mass, so we have to take an infinitesimal mass element of the rod and apply the Newton's law of gravitation between this element and the point mass m to get force between this small mass element and point mass m.
We can then integrate this force to find the net force between the mass m and the rod.

Consider a differential element of mass $dM = \frac{M}{L}dx$ kept at a distance x from the point mass m.



Force between the point mass and the element is given by,

$$dF = G\frac{mdM}{x^2} = \frac{Gm\frac{M}{L}dx}{x^2}$$

Net force can be found by integrating the above expression,

$$F = \int dF = \frac{GMm}{L}\int_{a}^{a+L}\frac{dx}{x^2} = \frac{GMm}{a(a+L)}$$

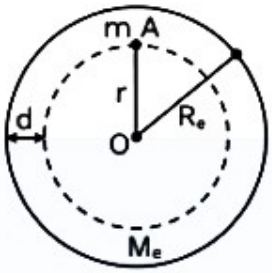
Acceleration Due to Gravity

A body accelerates when a force acts on it. For a body falling freely from a height towards the earth's surface, the velocity of the body increases in magnitude with time. We can say that the body is accelerated towards the centre of earth. This acceleration is known as acceleration due to gravity.

For a body outside the earth, total mass of the earth exerts a force on it.

For a body inside earth (in a tunnel or trench), only the mass of the sphere traced from that point exerts a force on it.

For a mass m kept at a depth d from the surface of earth as shown in the figure below,



Let us consider that the radius of earth is R_e and it is of uniform density 'ρ' whose mass is $M_e = \frac{4}{3}\pi R^3 \rho$.

Then, the mass of the sphere of radius r, $M_r = \frac{4}{3}\pi r^3 \rho$

The force on the mass m at A has magnitude,

$$\therefore \quad F = \frac{G\frac{4}{3}\pi r^3 \rho m}{r^2}$$

$$\therefore \quad F = Gm\left(\frac{4\pi\rho}{3}\right)r$$

$$\therefore \quad F = Gm\frac{4}{3}\pi \times \frac{3}{4\pi R_e^3} M_e \times r \qquad \ldots\left[\because M_e = \frac{4}{3}\pi R_e^3 \rho\right]$$

$$\therefore \quad F = \frac{GM_e m}{R_e^3} r \qquad \ldots(ii)$$

If the mass m is kept at the surface of earth, then $r = R_e$

$$\therefore \quad F = \frac{GM_e m R_e}{R_e^3}$$

$$\therefore \quad F = \frac{GM_e m}{R_e^2} \qquad \ldots(iii)$$

By Newton's Second law,

$F = ma = mg$

Because the acceleration experienced by the mass m is usually denoted by the symbol g,

Thus, $mg = \frac{GM_e m}{R_e^2}$

$\therefore \quad g = \frac{GM_e}{R_e^2}$

On putting the values,

$M_e = 6 \times 10^{24}$ kg

$R_e = 6.4 \times 10^{6}$ m

$G = 6.67 \times 10^{-11}$ Nm^2/kg^2

We get, $g = \frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{(6.4 \times 10^6)^2}$

$g = 9.8$ m/s^2

Q. **What is the value of the acceleration due to gravity on the surface of a pulsar of mass $M = 1.98 \times 10^{30}$ kg and radius R = 12 km rotating with time period T = 0.041 seconds.**
($G = 6.67 \times 10^{-11}$ MKS)

Sol. Here, $G = 6.67 \times 10^{-11}\ N\ m\ /\ kg^2$

Mass of the pulsar, $M = 1.98 \times 10^{30}$ kg

Radius of the pulsar, $R = 12$ km $= 12 \times 10^3$ m

Acceleration due to gravity on the surface of the pulsar is

$$g = \frac{GM}{R^2}$$

Substituting the given numerical values, we get

$$g = \frac{(6.67 \times 10^{-11} Nm^2\ /\ kg^2)(1.98 \times 10^{30}\ kg)}{(12 \times 10^3 m)^2}$$

$$= 0.092 \times 10^{13}\ m\ /\ s^2 = 9.2 \times 10^{11}\ m\ /\ s^2$$

Q. **Find the relation between the acceleration due to the gravity on the surface of the earth and on a planet respectively, whose mass and radius are twice that of the earth?**

Sol. Let the acceleration due to gravity of that planet be g_p

As, $$g_p = \frac{GM_p}{R_p^2}$$

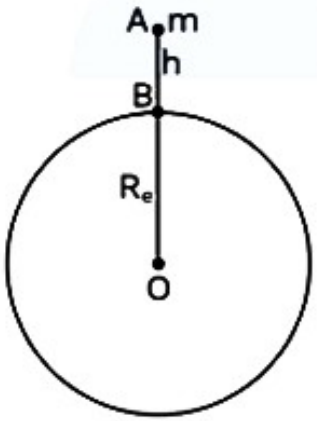
According to the question
$M_p = 2M_e$ and $R_p = 2R_e$

$$\therefore \quad g_p = \frac{G2M_e}{4R_e^2}$$

$$\therefore \quad g_p = \frac{1}{2}g_e \quad \ldots\left[\because g_p = \frac{GM_e}{R_e^2}\right]$$

Acceleration due to gravity above the surface of earth

Consider a point mass m kept at a height h above earth's surface.



The magnitude of force on the point mass m will be

$$F_h = \frac{GM_em}{(R_e + h)^2}$$

The acceleration of the point mass will be

$$g_h = \frac{F_h}{m} = \frac{GM_em}{(R_e + h)^2m}$$

$$\therefore \quad g_h = \frac{GM_e}{(R_e + h)^2} \quad \ldots(ii)$$

Above equation can be re-written as,

$$g_h = \frac{GM_e}{[R_e + h]^2} = \frac{GM_e}{R_e^2\left[1+\frac{h}{R_e}\right]^2}$$

$$\therefore \quad g_h = g\left[1+\frac{h}{R_e}\right]^{-2}$$

If $h << R_e$, then we can use binomial theorem to write,

$$\therefore \quad g_h = g\left[1-\frac{2h}{R_e}\right] \qquad \left\{\because (1+x)^n = 1+nx \text{ when } x << 1\right\}$$

Q. **The height above the surface of Earth at which the acceleration due to gravity decreases by 20% is (Take Earth's radius = R).**

Sol. If g and g_h denote the acceleration due to gravity at the Earth's surface and at the height h above the Earth's surface respectively, then

$$\frac{g_h}{g} = \frac{R^2}{(R+h)^2}$$

But according to the question

$$g_h = g - \frac{20}{100}g = \frac{80}{100}g = \frac{4}{5}g$$

$$\therefore \frac{4}{5} = \frac{R^2}{(R+h)^2}$$

or $$\sqrt{\frac{4}{5}} = \frac{R}{(R+h)} \Rightarrow \frac{2}{\sqrt{5}} = \frac{R}{R+h} \Rightarrow 2R + 2h = \sqrt{5}R$$

or $$h = \frac{\left(\sqrt{5}-2\right)R}{2} = \left(\frac{\sqrt{5}}{2}-1\right)R$$

Q. **Find the height above Earth's surface at which acceleration due to gravity becomes half of its value at the Earth's surface.**

Sol. It is given that $g_h = g\left[\frac{R}{R+h}\right]^2$

$$\frac{g_h}{g} = \left[\frac{R}{R+h}\right]^2$$

$$\therefore \quad \frac{1}{\sqrt{2}} = \left[\frac{R}{R+h}\right]$$

$$\therefore \quad R + h = \sqrt{2}R$$

$$h = (\sqrt{2} - 1)R$$

$$\therefore \quad h = (1.414 - 1)\ 6.4 \times 10^6$$

$$\therefore \quad h = 2.65 \times 10^6 \text{ m}$$

Acceleration due to gravity is reduced to half of its value on the Earth's surface at a height 2.65×10^6m above Earth's surface.

Acceleration due to gravity below the surface of earth

Consider a point mass m kept at depth d from the Earth's surface as shown.

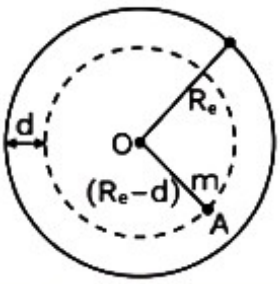


Fig: A point mass m at depth d from earth's surface

The gravitational force acting on the point mass m is only due to the smaller sphere (shown by dotted line) formed inside Earth.

Mass of the smaller sphere, $M_s = \frac{4}{3}\pi(R_e - d)^3\rho$ and

Total mass of Earth, $M_e = \frac{4}{3}\pi R_e^3\rho$

$$\therefore \quad \frac{M_s}{M_e} = \left[\frac{R_e - d}{R_e}\right]^3$$

By law of gravitation,

Force acting on the point mass, $F_d = \frac{GM_s m}{(R_e - d)^2}$

$\therefore \quad F_d = GM_e m \frac{(R_e - d)}{R_e^3}$

Also, by Newton's second law,

$a = \frac{F}{m}$

$\therefore \quad g_d = \frac{F_d}{m}$

$\therefore \quad g_d = GM_e \frac{(R_e - d)}{R_e^3}$

$\therefore \quad g_d = g \frac{R_e - d}{R_e} \quad \ldots \left[\because g = \frac{GM_e}{R_e^2}\right]$

$\therefore \quad g_d = g\left[1 - \frac{d}{R_e}\right]$

So, we can see that the acceleration due to gravity decreases on going below Earth's surface.

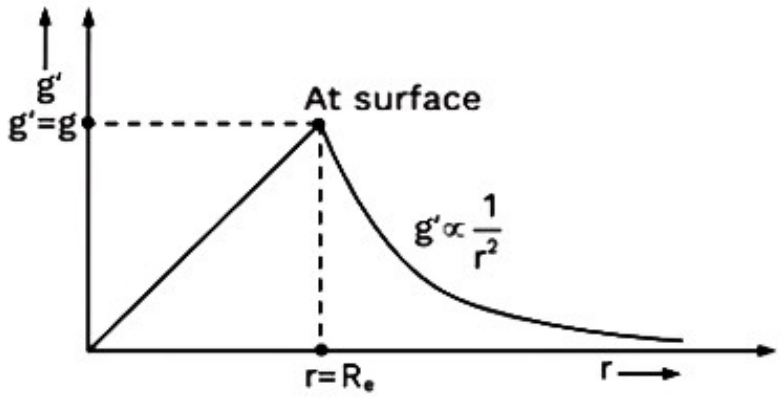
Note : The acceleration due to gravity is maximum at the earth's surface. It decreases either we go at higher altitudes or we move below the surface.

At centre $d = R_e$

$\therefore \quad g_d = g\left[1 - \frac{R_e}{R_e}\right] = g[1 - 1]$

$\therefore \quad g_d = 0$

Acceleration in a body due to earth's gravity at centre of Earth is zero. The variation of acceleration due to gravity with the distance from the centre of the earth may be drawn graphically as



Q. At what depth the acceleration due to gravity becomes zero?

Sol. As $g' = g\left[1-\frac{d}{R_e}\right]$

$\therefore \quad g\left[1-\frac{d}{R_e}\right] = 0$

$1-\frac{d}{R_e} = 0$

$\therefore \quad 1 = \frac{d}{R_e}$

$\therefore \quad d = R_e$

Acceleration due to gravity becomes zero at the centre of Earth.

Q. At what depth from the surface of the earth, the acceleration due to gravity will be $\frac{1}{4}$th the value of g on the surface of the earth?

Sol. As $g' = g\left[1-\frac{d}{R_e}\right]$

Given that,

$g' = \frac{g}{4}$

$\therefore \quad \frac{g}{4} = g\left[1-\frac{d}{R_e}\right]$

$\therefore \quad \frac{1}{4} = 1 - \frac{d}{R_e}$

$\therefore \quad \frac{d}{R_e} = 1-\frac{1}{4}$

$\therefore \quad d = 0.75\ R$

Q. **Some scientists travel to a far away planet in space whose radius R is one tenth of the radius of Earth but the mass density is same as that of Earth. The team of scientists dugs a tunnel of depth R/5 on its surface and start lowering a thin wire of same length R/5 having a linear mass density 0.001 kg/m in the tunnel. If the wire is not touching anywhere, then find the force acting at the top end by which a person is holding this wire in the tunnel. (Take radius of Earth= 6×10^6 m and $g_{Earth} = 10\ m/s^2$)**

Sol. Gravitational field inside the planet at a distance r from centre,

$$g_i = \frac{GM}{R^3} r = \frac{G\left(\frac{4}{3}\pi R^3\right)}{R^3} r = \frac{4}{3} G\pi r\rho$$

Force applied by the person on the wire at rest is the weight of the wire. Since, gravitational field is variable so force on each part of the wire is different. Consider a small portion dr of the wire at a distance r from centre of the planet, so its weight will be

$$dW = (dm)g_i$$

$$= (\lambda dr)\left(\frac{4}{3} G\pi r\rho\right) = \left(\frac{4}{3} G\pi\rho\right)(\lambda r dr)$$

Net weight, $W = \int dW = \left(\frac{4}{3} G\pi\rho\lambda\right) \int_{\left(\frac{4R}{5}\right)}^{R} r dr = \left(\frac{4}{3} G\pi\rho\lambda\right)\left[\frac{r^2}{2}\right]_{\left(\frac{4R}{5}\right)}^{R}$

$$W = \left(\frac{4}{3} G\pi\rho\lambda\right)\left(\frac{9}{50}\right) R^2$$

Density of the earth, $\rho = \frac{M_E}{\frac{4}{3}\pi R_E^3}$, Also, $R = \frac{R_E}{10}$

Putting the values of ρ and R in equation (i), we get

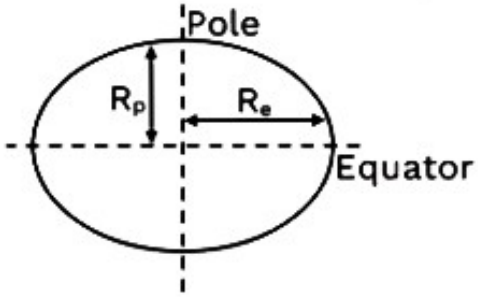
$$W = \left(\frac{9}{5 \times 10^3}\right) g_E \lambda R_E$$

$$= \frac{9}{5 \times 10^3} \times 10 \times 10^{-3} \times 6 \times 10^6 = 108\,N$$

So, net force applied by the person to hold the wire = 108 N

Variation of g due to the shape of the Earth

The real shape of Earth is not exactly spherical but we assume it to be spherical in most of our problems. It has actually an oval shape or we can say that it is bulged at the equator. Its equatorial radius (R_e) is higher than its polar radius (R_p). (Approximately, $R_e = R_p + 21$ km)

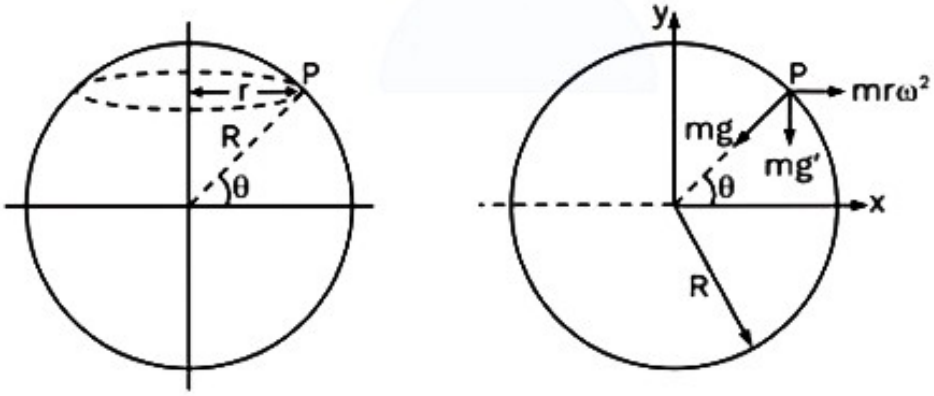


hence, $g_p = \frac{GM}{R_p^2}$ and $g_e = \frac{GM}{R_e^2}$

where g_p and g_e are acceleration due to gravity on pole and equator.
Hence, $g_p > g_e$

Effect of rotation of Earth on value of g

For a mass m kept at a general point P on Earth's surface, a pseudo force is acting on it from the rotating frame of reference as shown in the figure.
The weight of the mass m is towards the centre of Earth.



Hence, mg' will be the resultant of forces mg and $mr\omega^2$ and will be directed slightly away from the centre. The resultant mg' will be,

$$mg' = \sqrt{(mg)^2 + (mr\omega)^2 + 2(mg)(mr\omega)^2 \cos(180° - \theta)}$$

$$\Rightarrow \quad g' = \sqrt{g^2 + r^2\omega^4 + 2gr\omega^2 \cos(180° - \theta)}$$

$$\text{or} \quad g' = \sqrt{g^2 + r^2\omega^4 - 2gr\omega^2 \cos\theta}$$

Ignoring $r^2\omega^4$ (as it is too small compared to other terms) and putting $r = R\cos\theta$, we get

$$g' = (g^2 - 2gR\omega^2 \cos^2\theta)^{1/2}$$

$$= g\left(1 - \frac{2R\omega^2 \cos^2\theta}{g}\right)^{1/2}$$

$$= g\left(1-\frac{R\omega^2\cos^2\theta}{g}\right) \qquad \text{(by binomial approximation)}$$

$\Rightarrow \quad g' = g - R\omega^2\cos^2\theta$

Now,

1. At equator $\theta = 0°$ and $g' = g - R\omega^2$
2. At pole $\theta = 90°$ and $g' = g$
3. Fractional change in g at latitude $\theta = \dfrac{g'-g}{g} = \dfrac{-R\omega^2\cos^2\theta}{g}$

Q. **Weight of a man at equator is W if Earth does not rotate. With what speed should Earth be rotating about its axis so that the weight of the man at equator is 0.75W?**

Sol. At the equator,

$$g_e = g - \omega^2 R\cos^2 0°$$

$$\frac{3}{4}g = g - \omega^2 R\cos^2 0° \text{ or } \omega^2 R = \frac{g}{4}$$

$$\omega = \sqrt{\frac{g}{4R}} = \sqrt{\frac{10}{4\times 6400\times 10^3}} = 0.63\times 10^{-3}\ \text{rad/s}$$

Q. **Find the percentage change in the acceleration due to gravity when a small body is taken to**

(a) An altitude of $\dfrac{R}{100}$

(b) At a depth of $\dfrac{R}{100}$

Sol. We know

(a) At an altitude h (< < R),

$$g = g_0\left(1-\frac{2h}{R}\right) \Rightarrow \frac{\Delta g}{g_0}\times 100 = \frac{-2h}{R}\times 100 = \frac{-2}{R}\times\frac{R}{100}\times 100\% = -2\%$$

(b) At a depth h

$$g < g_0\left(1-\frac{h}{R}\right) \Rightarrow \frac{\Delta g}{g_0} = -\frac{h}{R}\times 100 = -\frac{1}{R}\times\frac{R}{100}\times 100\% = -1\%$$

Gravitational Field

Gravitational force is a non-contact force. A mass produces a gravitational field around it and when another body is brought into this field, it experiences a force.
When a particle of mass m is placed in a gravitational field of intensity $\vec{I}$, then the force experienced by the particle is

$$\vec{F}_g = m\vec{I} \quad \Rightarrow \quad \vec{I} = \frac{\vec{F}_g}{m}$$

Gravitational field intensity (or gravitational field) at a point in space is defined as the gravitational force experienced by a test mass placed at that point divided by its mass.
Gravitational field vector is always towards the mass. For spherical bodies, the gravitational field is a radial vector.

Gravitational Field Due to different Bodies

Point Mass

To find the gravitational field due to a point mass, we keep a test mass m_0 at a distance r from the point mass m.
The gravitational force experienced by the mass at P is given as



Now, the gravitational force acting on the test mass due to the given point mass will be,

$$\vec{F} = \frac{Gmm_0}{r^2}(-\hat{r})$$

$$\vec{I} = \frac{\vec{F}}{m_0}$$

So, $\vec{I} = \frac{-Gm}{r^2}\hat{r}$

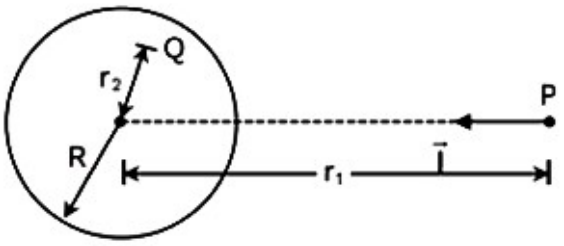
So, the gravitational field intensity at a distance r from a particle of mass m has a magnitude $\frac{GM}{r^2}$ and is directed "radially inwards".

Shell Theorem

The shell theorem provides simplifications that can be applied to objects placed inside or outside a spherically symmetrical body in gravitation as well as in electrostatics.
Two major outcomes from Shell theorem which are useful in our study are given below:

1. A solid uniform sphere, having mass m, produces the gravitational field similar to that produced by a point mass placed at the centre.
2. When the point of consideration lies outside a hollow, uniform sphere, the gravitational field intensity is similar to that produced by a point mass.
   When the point of consideration lies inside a uniform spherical shell (hollow sphere), the gravitational field intensity is zero due to the shell.

The following figure describes the above theorem



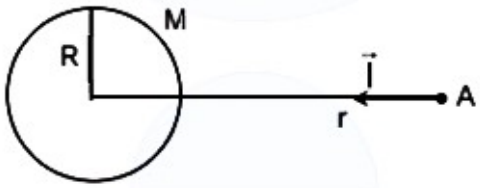
At point P, $I = -\frac{GM}{r_1^2}\hat{r}$

At point Q, $|\vec{I}| = 0$.

Uniform Thin Spherical Shell

At an external point (r > R)

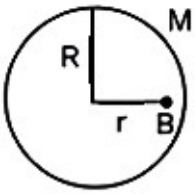
As per shell theorem, the gravitational field intensity is similar to that produced by a point mass of mass M assumed to be placed at centre of shell.



So, $\boxed{\vec{I} = -\frac{GM}{r^2}\hat{r}}$ at A

At an internal point (r < R)

As per shell theorem, gravitational field intensity is zero due to the shell.

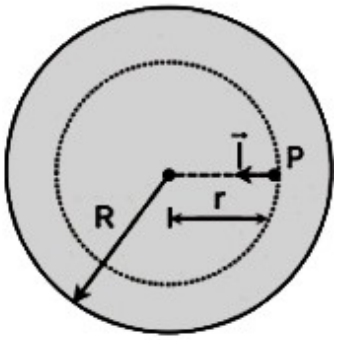


$\boxed{|\vec{I}| = 0}$ at B.

Uniform Solid Sphere

For a point outside the sphere, i.e., at a radial distance r (> R) from the centre,

$$\vec{I} = -\frac{GM}{r^2}\hat{r} \qquad (r > R)$$



For a point inside the sphere,

The gravitational field intensity at point P, is due to a solid sphere of radius r only. Let the mass of this sphere of radius r is m', then,

$$m' = \frac{m}{\frac{4}{3}\times\pi R^3}\times\frac{4}{3}\pi r^3 = \frac{mr^3}{R^3}$$

The field intensity is

$$\vec{I} = -\frac{Gm'}{r^2}\hat{r}$$

Or $\vec{I} = -\frac{Gmr}{R^3}\hat{r}$ $\qquad$ (using $m' = \frac{mr^3}{R^3}$)

So, the gravitational field due to a solid sphere is,

$$I = \begin{cases} -\frac{Gmr}{R^3}\hat{r}\ , & 0 < r \le R \\ -\frac{Gm}{r^2}\hat{r}\ , & r \ge R \end{cases}$$

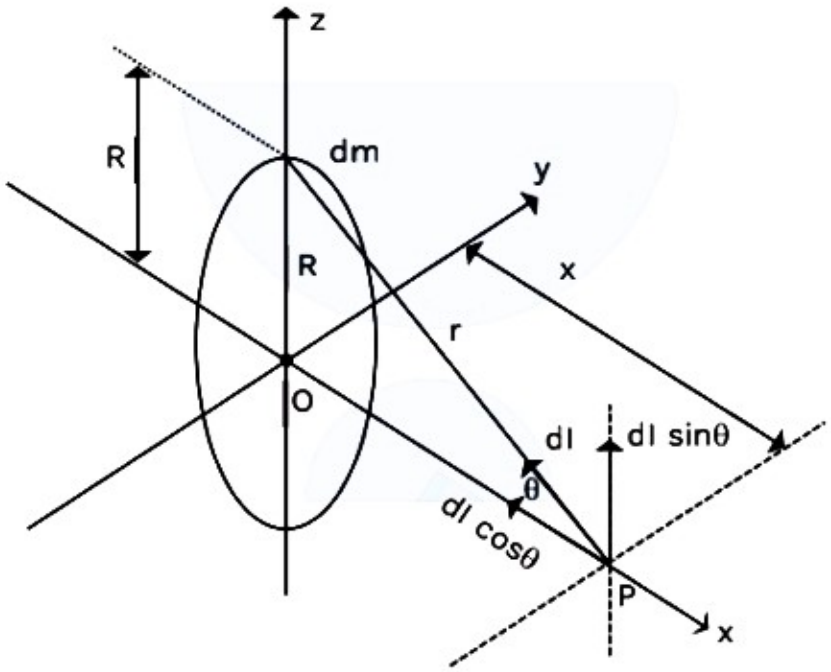
Uniform Ring

To find the gravitational field on the axis of a uniform ring of mass m and radius R, consider an infinitesimal mass element dm on the periphery of the ring as shown in the figure.

This infinitesimal mass element has two components of gravitational field at the point P on the axis of the ring as shown.

For such an element, there is always a diametrically opposite element whose dI $\sin\theta$ field component cancels the same component of the first element.

For all the mass components, the dI $\sin\theta$ component cancels out while the dI $\cos\theta$ components add up.



Net gravitational field at P, $I = \int dI_x = \int dI\cos\theta = \int \frac{Gdm}{r^2} \times \frac{x}{r} = \frac{Gx}{r^3}\int dm$

$$\Rightarrow \quad I = \frac{Gmx}{r^3}$$

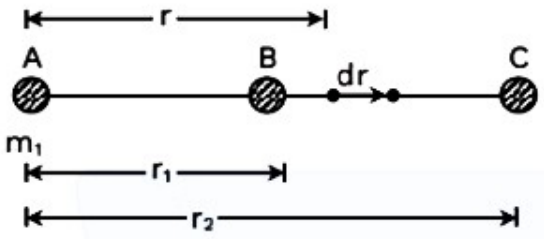
$$\Rightarrow \quad \vec{I} = -\frac{Gmx}{(R^2 + x^2)^{3/2}}\hat{i}$$

Gravitational Potential Energy

Change in potential energy of a system is given by negative of the work done by the conservative force.

$$U_f - U_i = -\int_i^f \vec{F}.\overrightarrow{dr} \qquad [f \rightarrow \text{final, } i \rightarrow \text{initial}]$$

For a system of two masses m_1 and m_2 where m_1 is kept fixed at a point A and m_2 is moved from point B to point C very slowly,



For a small displacement dr of the mass m_2, external work done is given by,

$$dW_{ex} = \vec{F}_{ext} \cdot d\vec{r}$$

As the mass m_2 is being moved very slowly, so $|\vec{F}_{ext}| = |\vec{F}_G| = \frac{Gm_1m_2}{r^2}$

Then work done by the gravitational force in the displacement dr is given by,

$$dW = -\frac{Gm_1m_2}{r^2}dr$$

Change in potential energy of the masses,

$$dU = -dW = \frac{Gm_1m_2}{r^2}dr$$

or $$U(r_2) - U(r_1) = \int dU$$

$$= \int_{r_1}^{r_2} \frac{Gm_1m_2}{r^2}dr$$

$$= Gm_1m_2\left[-\frac{1}{r}\right]_{r_1}^{r_2}$$

$$\Rightarrow \quad U(r_2) - U(r_1) = Gm_1m_2\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

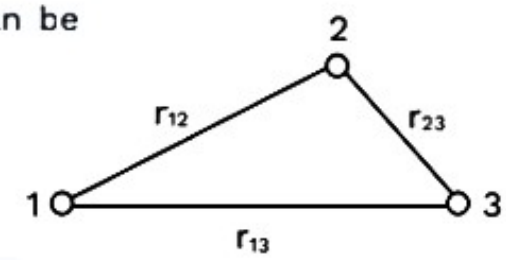
In general, we choose $r_1 = \infty, U(r_1) = 0$. (infinity as the reference point)

$$\therefore \quad U(r) - U(\infty) = Gm_1m_2\left(\frac{1}{\infty} - \frac{1}{r}\right)$$

$$\Rightarrow \quad U = -\frac{Gm_1m_2}{r}$$ (when particles are at a distance r from each other.)

For a system of three or more masses, the same result can be applied. We have to add up for all the possible pairs.

For example, consider the system of three particles shown,



$$U_{total} = U_{12} + U_{13} + U_{23}$$

$$= -G\left(\frac{m_1m_2}{r_{12}} + \frac{m_1m_3}{r_{13}} + \frac{m_2m_3}{r_{23}}\right)$$

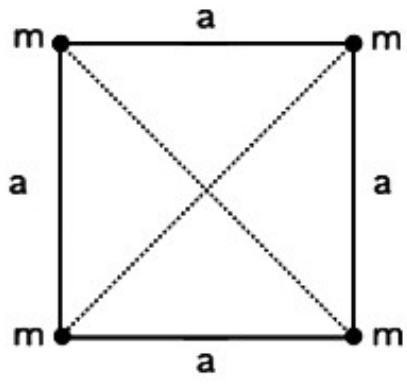
Similarly, for four particles system

$$U_{total} = U_{12} + U_{13} + U_{23} + U_{23} + U_{24} + U_{34}$$

Note : For a system of N particles, there will be $\frac{N(N-1)}{2}$ pairs.

Q. **Find the gravitational potential energy of a system of four point masses kept at the corners of a square of side length a.**

Sol. For a system of four point masses, there will be 6 pairs.
Four pairs are the masses at the edges and two pairs are the masses at the diagonal extremities.
The potential energy of the pairs at edges is



$$U_1 = 4 \times \left(\frac{-Gm^2}{a}\right)$$

The potential energy for the pairs at the diagonals is

$$U_2 = 2 \times \left(\frac{-Gm^2}{a\sqrt{2}}\right)$$

So, total potential energy is $U = \frac{-Gm^2}{a}\left(4 + \sqrt{2}\right)$

Q. **Two masses m_1 and m_2 are released from rest to move towards each other under mutual gravitational attraction. Initially they were large distance apart. Find their relative speed when they are at a separation r?**

Sol. Gravitational force acts as an internal force for the system of two masses, so the linear momentum and the total mechanical energy of the system remain conserved.

$$\sum p_i = \sum p_f$$

$$\Rightarrow \quad 0 = m_1v_1 - m_2v_2 \qquad \ldots(i)$$

Negative sign has been chosen to keep v_1 and v_2 as magnitude of velocities only.

Also, $U_i + K_i = U_f + K_f$

$$0 + 0 = \frac{-Gm_1m_2}{r} + \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \qquad \ldots(ii)$$

From (i), we get

$$m_1v_1 = m_2v_2 = p \text{ (say)}$$

From (ii), we get

$$\frac{Gm_1m_2}{r} = \frac{p^2}{2m_1} + \frac{p^2}{2m_2}\left(\text{as } k = \frac{p^2}{2m}\right)$$

$$\Rightarrow \quad p = m_1m_2\sqrt{\frac{2G}{(m_1 + m_2)r}}$$

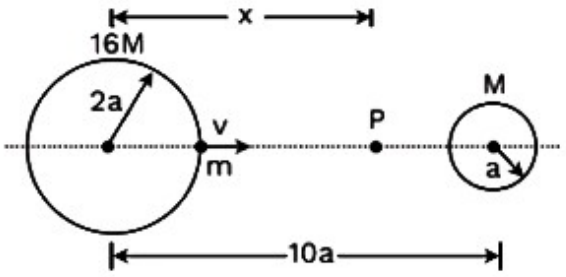
Now relative speed $= v_1 + v_2$

$$= \frac{p}{m_1} + \frac{p}{m_2}$$

$$= \sqrt{\frac{2G(m_1 + m_2)}{r}}$$

Q. **Two stars of masses 16M and M with radii 2a and a respectively are at a distance 10a apart. If a body of mass m is fired from the surface of the bigger star, then what should be the minimum firing speed to make it reach the surface of the smaller star?**

Sol. For minimum possible firing speed, we have to project the mass m from the surface of bigger star with such an energy that it reaches the point P between the stars where net gravitational field is zero. After that it will be simply attracted by the gravitational field of the smaller star.



Net gravitational field at P should be zero, so

$$\frac{16GMm}{x^2} = \frac{GMm}{(10a - x)^2}$$

$$\Rightarrow \quad \left(\frac{10a - x}{x}\right)^2 = \frac{1}{16}$$

$$\Rightarrow \quad \frac{10a - x}{x} = \frac{1}{4}$$

$$\Rightarrow \quad x = 8a$$

Using energy conservation between projection point and point P,

$$-\frac{16GMm}{2a} + \frac{1}{2}mv^2 - \frac{GMm}{8a} = -\frac{16GMm}{8a} - \frac{GMm}{2a}$$

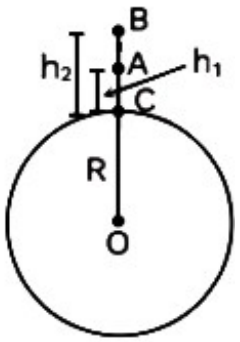
$$\Rightarrow \quad \frac{1}{2}v^2 - \frac{GM}{a}\left(2 + \frac{1}{2} - \frac{1}{8} - 8\right)$$

$$\Rightarrow \quad v = \frac{3}{2}\sqrt{\frac{5GM}{a}}$$

GRAVITATIONAL POTENTIAL ENERGY OF AN OBJECT IN THE FIELD OF EARTH

Consider two points A and B at heights h_1 and h_2 respectively, above the surface of Earth.

In moving a mass m from A to B, the external work done is given by,



W_{AB} = Force × displacement

$= mg(h_2 - h_1)$ ...(i)

Lets say that potential energy at point A is U_A and the potential energy at point B is U_B, then

We can write, $U_B - U_A = mg(h_2 - h_1)$

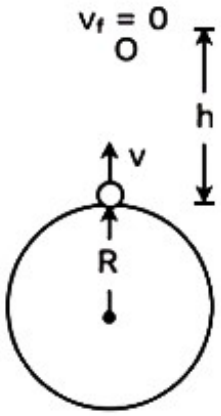
It can also be said that the work done in moving the particle is just the difference of potential energy between its final and initial positions.

Height achieved by a body projected from the surface of the Earth with very high velocity

Consider a body of mass m to be projected from ground with speed v in vertically upward direction. Let the body achieves a maximum height h.

At the maximum height, the kinetic energy of the body becomes zero.

By conservation of energy,



$K_i + U_i = K_f + U_f$

$$\Rightarrow \quad \frac{1}{2}mv^2 - \frac{GMm}{R} = 0 - \frac{GMm}{R+h}$$

$$\Rightarrow \quad v^2 = \frac{2GM}{R} - \frac{2GM}{R+h} = \frac{2GM}{R}\cdot\frac{h}{(R+h)} = v_e^2\frac{h}{R+h} \qquad \left[\because v_e = \sqrt{\frac{2GM}{R}}\right]$$

$$\Rightarrow \quad \frac{R+h}{h} = \frac{v_e^2}{v^2} = \frac{1}{k^2} \qquad \left[\text{If } v = kv_e \text{ or } k = \frac{v}{v_e} < 1\right]$$

$$\Rightarrow \quad 1 + \frac{R}{h} = \frac{1}{k^2} \qquad \Rightarrow \frac{R}{h} = \frac{1-k^2}{k^2}$$

$$\Rightarrow \quad \text{Maximum height, } h = \frac{Rk^2}{1-k^2}$$

Escape Velocity

Escape velocity is the minimum velocity given to a body at the surface of a planet so that the body leaves the gravitational field of the planet and reaches infinity.

Consider that a body of mass m is projected from the earth's surface with speed v_e.

The total mechanical energy of the body will remain conserved.

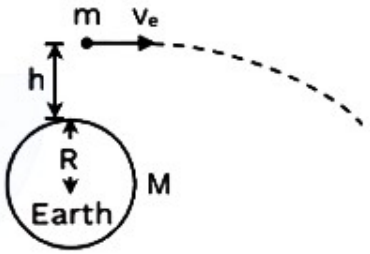
So, Total mechanical energy on earth = Total mechanical energy at infinity

$$K_i + U_i = K_f + U_f$$

or, $$\frac{1}{2}mv_e^2 - \frac{GMm}{R} = 0 + 0$$

$$v_e = \sqrt{\frac{2GM}{R}}$$

On the earth's surface, on putting the values of G and M=Mass of earth, and R= Radius of earth, we get



$K_f = K_{min} = 0$ (Velocity at ∞ is zero)

$U_f = U_\infty = 0$

Escape velocity, v_e = 11.2 km/s

If the body is projected from a height h above the surface,

$$v_e = \sqrt{\frac{2GM}{R+h}}$$

Q. **What is the change in gravitational potential energy of a body of mass m when it is raised to a height nR from the surface of earth (R= Radius of earth)**

Sol. Gravitational potential energy of mass m at any point at a distance r from centre of earth is

$$U = -\frac{GMm}{r}$$

At the surface of earth r = R,

$$\therefore U_s = -\frac{GMm}{R} = -\frac{GMmR}{R^2} = -mgR \quad \left(\because g = \frac{GM}{R^2}\right)$$

At the height h = nR from the surface of earth

$$\therefore U_h = -\frac{GMm}{R(n+1)} = -\frac{mgR}{(n+1)}$$

Change in gravitational potential energy is

$$\Delta U = U_h - U_s = -\frac{mgR}{(n+1)} - (-mgR)$$

$$= -\frac{mgR}{(n+1)} + mgR = mgR\left(1 - \frac{1}{(n+1)}\right) = mgR\left(\frac{n}{n+1}\right)$$

Q. **The ratio of the masses and radii of two planets are 2 : 3 and 4 : 9. What is the ratio of the escape speed at their surface?**

Sol. $\frac{V_1}{V_2} = ?, \frac{M_1}{M_2} = \frac{2}{3}$ and $\frac{R_1}{R_2} = \frac{4}{9}$

As we know $V = \sqrt{\frac{2GM}{R}}$

$$\therefore \quad \frac{V_1}{V_2} = \sqrt{\frac{M_1}{M_2} \times \frac{R_2}{R_1}}$$

$$= \sqrt{\frac{2}{3} \times \frac{9}{4}} = \sqrt{\frac{3}{2}}$$

$$\therefore \quad V_1 : V_2 = \sqrt{3} : \sqrt{2}$$

Q. **What will be the escape speed from a planet having mass 8 times that of earth and diameter 4 times that of the earth? [v_e = 11.2 km/s]**

Sol. $v_p = ?$

Now, $v = \sqrt{\frac{2GM}{R}}$

$\therefore \quad \frac{v_p}{v_e} = \sqrt{\frac{M_p}{M_e} \times \frac{R_e}{R_p}} = \sqrt{\frac{8M_e}{M_e} \times \frac{R_e}{4R_e}}$

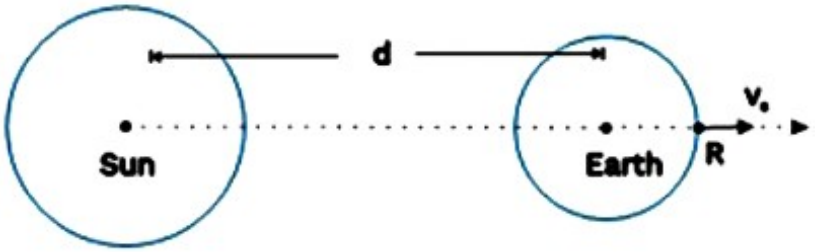
$\therefore \quad v_p = \sqrt{2}\, v_e$

$= \sqrt{2} \times 11.2$

$= 15.84$ km/s

Q. **Along the line segment joining the Earth and the Sun, a rocket is launched perpendicular to the Earth's surface. Find the minimum initial velocity given to the rocket so that it leaves the Earth-Sun system.**
Consider Sun to be 3×10^5 times heavier than Earth and the separation between Sun and Earth to be 2.5×10^4 times the radius of Earth. Also take the escape velocity from Earth's gravitational field to be 11.2 km/s.

Sol.



Given: Mass of the sun, $M_s = 3 \times 10^5 \times M_e$

Distance between Earth and the Sun, $d = 2.5 \times 10^4 R_e$,

Escape speed from Earth's field, $v_e = 11.2 \text{km s}^{-1}$

$\therefore v_e = \sqrt{\frac{2GM}{R_e}} = 11.2 \text{km s}^{-1}$

Let the minimum velocity needed by the rocket to escape the Earth – Sun system = v_s

By conservation of energy,

$$\frac{1}{2}mv_s^2 = \left(\frac{GM_em}{R_e} + \frac{GM_sm}{d+R_e}\right) = \left(\frac{GM_em}{R_e} + \frac{G\times 3\times 10^5 M_em}{(2.5\times 10^4 R_e + R_e)}\right)$$

$$\frac{1}{2}mv_s^2 = \frac{GM_em}{R_e} + \frac{G\times\left(3\times 10^5\right)M_em}{(2.5\times 10^4 R_e)} \qquad \left(\because 2.5\times 10^4 >> 1\right)$$

$$\therefore v_s^2 = \frac{2GM_e}{R_e} + \frac{24GM_e}{R_e}$$

$$\Rightarrow v_s^2 = v_e^2 + 12v_e^2 = 13v_e^2$$

$$\text{or } v_s = \sqrt{13}\, v_e = 40.38\,\text{km s}^{-1} \approx 42\,\text{km s}^{-1}$$

Gravitational Potential (V)

Gravitational potential energy per unit mass of a particle placed at a point is known as the Gravitational potential of that point.

It is equal to the work done in bringing a unit mass from infinity to a point in given gravitational field without acceleration.

If U_A and U_B denote potential energy at A and at B respectively, then change in potential per unit mass as it moves from A to B is

$$V_B - V_A = \frac{U_B - U_A}{m} \text{ (J / kg)}$$

If A is taken as a reference point where potential energy is taken as zero,

$V_A = 0 \Rightarrow U_A = 0.$

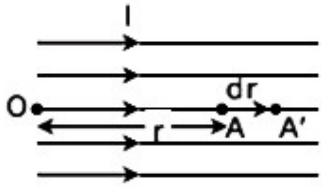
Let $V_B = V$ and $U_B = U.$

Then, $V = \frac{U}{m}$

Relation Between Gravitational Field and Potential

Force acting on a mass m in field $\vec{I}$, is

$$\vec{F} = m\vec{I}$$



Work done in displacing the mass from $\vec{r}$ to $\vec{r} + \overrightarrow{dr}$,

$$dW = \vec{F}.\overrightarrow{dr}$$

We know that the work done by conservative force is equal to negative of change in potential energy, so

$$dU = -dW$$

$$\Rightarrow \quad dU = -m\vec{I}.\overrightarrow{dr}$$

$$\Rightarrow \quad dV = \frac{dU}{m} = -\vec{I}.\overrightarrow{dr}$$

So, $dV = -\vec{I}.\overrightarrow{dr}$

or $V(r) = -\int_{\infty}^{r} \vec{I}.\overrightarrow{dr}$ ($r = \infty$ is taken as reference point where $V = 0$)

If $\vec{I} = I_x\hat{i} + I_y\hat{j} + I_z\hat{k}$ and $\overrightarrow{dr} = dx\hat{i} + dy\hat{j} + dz\hat{k}$, then

1. $V(r) = -\int_{\infty}^{x} I_x dx - \int_{\infty}^{y} I_x d_y - \int_{\infty}^{z} I_z d_z$

2. $I_x = -\frac{\partial V}{\partial x}, I_y = \frac{-\partial V}{\partial y}$ and $I_z = -\frac{\partial V}{\partial z}$ or $\vec{I} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$

Note : $\frac{\partial V}{\partial x}$ or partial differentiation of V with respect to x means differentiating V with respect to x treating y and z as constants.

Q. **A particle is released from a point having a gravitational potential of 5 J/kg. It passes a point B with a speed of 1 m/s after some time. If mass of the particle is 1 kg, then find the gravitational potential of the point B.**

Sol. The particle is released from rest and it has a velocity of 1 m/s at the point B. So, the kinetic energy of the particle increases on moving from A to B. Thus, the potential energy must have decreased.

As per law of conservation of mechanical energy,

$$K_A + U_A = K_B + U_B$$

$$0 + U_A = \frac{1}{2} \times 1 \times (1)^2 + U_B$$

$\Rightarrow \quad U_A - U_B = 0.5$

Divide above equation by the mass of the particle,

$$\frac{U_A - U_B}{m} = \frac{0.5}{1} J/kg$$

$\Rightarrow \quad V_A - V_B = 0.5$ J/kg

$\Rightarrow \quad V_B = 5 - 0.5 = 4.5$ J/kg

Q. **The gravitational field due to a mass distribution is given by $\vec{I} = \frac{k}{x^2}\hat{i}$, where k is a constant. Assuming the potential to be zero at infinity, find the potential at a point x = a.**

Sol. We know,

$$dV = -\vec{I}.\overrightarrow{dr}$$

$$\Rightarrow \quad dV = -\frac{k}{x^2}dx$$

$$\Rightarrow \quad \int dV = -k\int \frac{dx}{x^2}$$

$$\Rightarrow \quad V = \frac{k}{x} + C$$

When $x = \infty$, $V = 0 \Rightarrow C = 0$

$$\Rightarrow \quad V = \frac{k}{x}$$

$$\therefore \quad \text{At } x = a, \ V = \frac{k}{a}$$

Q. **The variation of the gravitational potential in a region is given by $V = \frac{8x}{x^2 + p^2}$.**

What is the value of the gravitational field at point x = p.

Sol. As V is function of x only, so, $\vec{E} = -\frac{\partial V}{\partial x}\hat{i}$

$$= -8\frac{d}{dx}\left(\frac{x}{x^2 + p^2}\right)\hat{i}$$

$$= -8\left(\frac{(x^2 + p^2) - x(2x)}{(x^2 + p^2)^2}\right)\hat{i}$$

$$\Rightarrow \quad \vec{E} = 8\frac{(x^2 - p^2)}{(x^2 + p^2)^2}\hat{i}$$

At $x = p$, $\vec{E} = 0$.

Q. **Considering the gravitational potential to be zero at $x = \infty$, find its value at a point x if the gravitational field is given by $I = kx^{-3/2}$ in x-direction.**

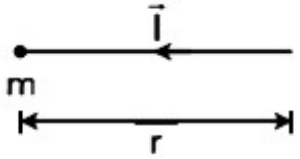
Sol. $V(x) = -\int_{\infty}^{x} I dx = -k\int_{\infty}^{x} x^{-3/2}dx = +2k\left[\frac{1}{\sqrt{x}}\right]_{\infty}^{x} = \frac{2k}{\sqrt{x}}$.

Gravitational Potential due to Different Bodies

Point Mass

Gravitational field due to a point mass is given by,

$$\vec{I} = -\frac{Gm}{r^2}\hat{r}$$



Now, from the definition of potential

$$dV = -\vec{I}.\overrightarrow{dr}$$

$$\Rightarrow \quad dV = -\left(\frac{-Gm}{r^2}\hat{r}\right).(dr\hat{r}) \quad ...(\because \overrightarrow{dr} = dr\hat{r})$$

Integrating within suitable limits,

$$\int_0^V dV = \int_\infty^r \frac{Gm}{r^2} dr$$

$$[V]_0^V = -Gm\left[\frac{1}{r}\right]_\infty^r$$

$$\Rightarrow \quad V = -\frac{Gm}{r}.$$

The negative sign in potential signifies that when a particle moves from infinite distance towards a particle of mass m, the work done by gravitational field is positive and the potential energy of the system decreases.

Uniform thin Spherical Shell

Potential at an external point

For a point outside a shell, the mass of the shell can be assumed to be concentrated at its centre. So, it can be taken as a point mass.
So, potential at a distance r is given by

$$\boxed{V(r) = -\frac{GM}{r}} \qquad (\text{For } r \geq R)$$

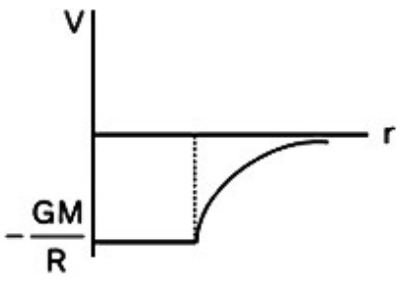
At r = R, $\boxed{V = -\frac{GM}{R}}$

Potential at an internal point

At all the points inside a spherical shell, the gravitational potential is same as the potential on the surface. It is so because gravitational field inside the shell is zero.
For r < R,

$$\boxed{V = -\frac{GM}{R}}$$

V-r graph for a spherical shell can be plotted as-



Uniform Solid Sphere

Gravitational field for a uniform solid sphere is given by,

$$I = \begin{Bmatrix} -\frac{GMr}{R^3}\hat{r}, 0 < r \le R \\ -\frac{GM}{r^2}\hat{r} \ , r \ge R \end{Bmatrix}$$

Now, to find the potential, following equation can be used-

$$dV = -\bar{I}.\overline{dr}$$

For a point outside the sphere :

$$dV = -\left(-\frac{GM}{r^2}\hat{r}\right).(dr\,\hat{r})$$

Integrating from r = ∞ to r = r, we get

$$\int_0^V dV = \int_\infty^r \frac{GM}{r^2}dr$$

$$\Rightarrow \quad V - 0 = GM\left[-\frac{1}{r}\right]_\infty^r \Rightarrow V = -\frac{GM}{r}$$

Potential at the surface is $V_s = -\frac{GM}{R}, r = R$

For a point inside the sphere :

$$dV = -\left(-\frac{GMr}{R^3}\hat{r}\right).(dr\,\hat{r})$$

$$dV = \frac{GM}{R^3}rdr$$

Integrating from r = R to r = r, we get

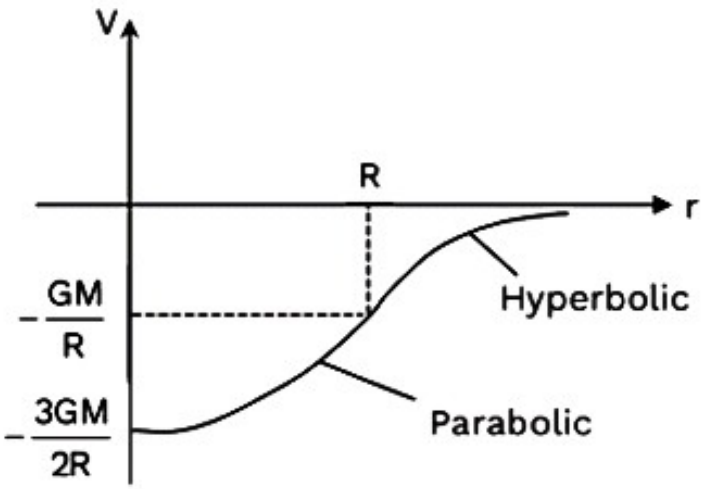
$$\int_{V_s}^V dV = \int_R^r \frac{GM}{R^3}rdr$$

$$\Rightarrow \quad V - V_S = \frac{GM}{R^3}\left[\frac{r^2}{2}\right]_R^r$$

$$\Rightarrow \quad V - V_S = \frac{GM}{R^3}\left(\frac{r^2 - R^2}{2}\right)$$

$$\text{or} \quad V = -\frac{GM}{R} + \frac{GM(r^2 - R^2)}{2R^3}$$

$$\text{or} \quad V = -\frac{GM}{2R^3}(3R^2 - r^2).$$



Uniform Ring

Gravitational field on the axis of a ring is given by,

$$\boxed{\vec{I} = \frac{-Gmx}{(R^2 + x^2)^{3/2}}\hat{i}}$$

To find potential, we can use,

$$dV = -\vec{I}.\vec{dr}$$

here, $\vec{dr} = dx\hat{i}$

$$\vec{I} = \frac{-Gmx}{(R^2 + x^2)^{3/2}}\hat{i}$$

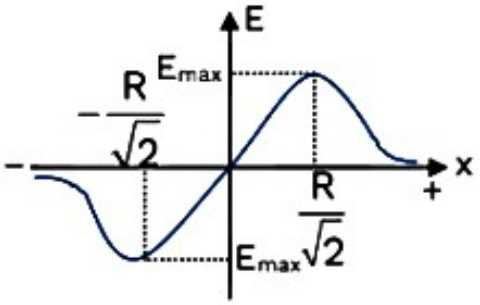
So, $dV = \dfrac{+Gmxdx}{(R^2 + x^2)^{3/2}}$

Integrating within limits,

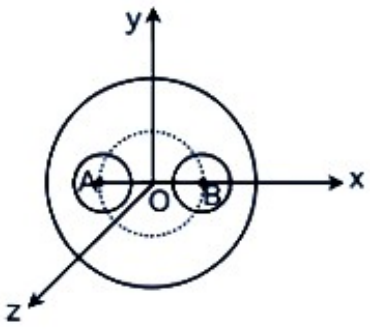
$$V = \int_{\infty}^{x} \frac{Gmxdx}{(R^2 + x^2)^{3/2}}$$

$$V = -\frac{Gm}{\sqrt{(R^2 + x^2)}}$$

Graph of E versus X for ring



Q. **A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates. Two spheres of equal radii (1 unit) with their centres at A(−2, 0, 0) and B(2, 0, 0) respectively, are taken out of the solid leaving behind spherical cavities as shown in the figure.**



Then,

(A) **the gravitational force due to this object at the origin is zero**

(B) **the gravitational force at point B(2, 0, 0) is zero**

(C) **the gravitational potential is the same at all points of circle $y^2 + z^2 = 36$**

(D) **The gravitational potential is the same at all points on the circle $y^2 + z^2 = 4$**

Sol. Since cavities A and B are symmetric about O, the gravitational field at O is zero. Hence force is zero (a is correct). But gravitational force at point B cannot be zero as mass distribution is as asymmetric about B.

$y^2 + z^2 = 36$ is a circle in y-z plane. Cavities A and B are symmetric about y-z plane.

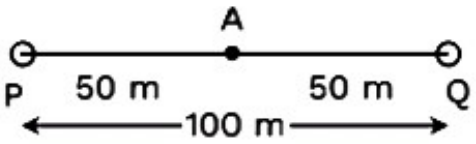
$\therefore$ Gravitational potential will be same at all the points on the circle. Same argument holds good for the circle $y^2 + z^2 = 4$.

Hence (A), (C) and (D) are correct.

Q. **Two spheres of masses 10^6 kg and 10^8 kg and each of radius 20 cm are separated by a distance of 100 m. What will be the potential at the mid-point of the line joining their centres?**

Sol. The total potential at A

$$V = V_P + V_Q = -\frac{Gm_P}{r} - \frac{Gm_Q}{r}$$



$$\therefore \quad V = -\frac{6.67 \times 10^{-11}}{50}[10^6 + 10^8]$$

$$V = -\frac{6.67 \times 10^{-11} \times 10^6 \times 101}{50} = -1.35 \times 10^{-4}\ \text{J/kg}$$

Q. **From a solid sphere of mass m and radius R, a spherical portion of radius $\frac{R}{2}$ is removed, as shown in the figure. Taking gravitational potential V = 0 at $r = \infty$, the potential at the centre of the cavity thus formed is (G = gravitational constant)**



Sol. Potential at point P (centre of cavity) before removing the spherical portion.

$$V_1 = \frac{-GM}{2R^3}\left(3R^2 - \left(\frac{R}{2}\right)^2\right)$$

$$= \frac{-GM}{2R^3}\left(3R^2 - \frac{R^2}{4}\right) = \frac{-11GM}{8R}$$



Mass of spherical portion to be removed, $M' = \frac{MV'}{V}$

$$= \frac{M\frac{4\pi}{3}\left(\frac{R}{2}\right)^3}{\frac{4\pi}{3}R^3} = \frac{M}{8}$$

Potential at point P due to spherical portion to be removed

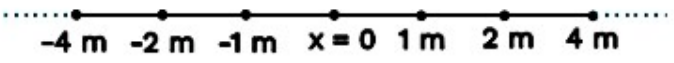
$$V_2 = \frac{-3GM'}{2R'} = \frac{-3G(M/8)}{2(R/2)} = \frac{-3GM}{8R}$$

$\therefore$ Potential at the centre of cavity formed

$$V_p = V_1 - V_2 = \frac{-11GM}{8R} - \left(\frac{-3GM}{8R}\right) = \frac{-GM}{R}$$

Q. **Infinite number of masses, each 1 kg, are placed along the x-axis at $x = \pm 1\,m, \pm 2\,m, \pm 4\,m, \pm 8\,m, \pm 16\,m$..... The magnitude of the resultant gravitational potential in terms of gravitational constant G at the origin (x = 0) is**

Sol.



Total gravitational potential at O is

$$V = 2\left[\frac{-G\times 1}{1} - \frac{G\times 1}{2} - \frac{G\times 1}{4} - \frac{G\times 1}{8} - \frac{G\times 1}{16} \ldots\ldots\right]$$

$$= -2G\left[\frac{1}{1} + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \ldots.\right]$$

$$= -2G\left[\frac{1}{1-\frac{1}{2}}\right] = -4G$$

So, the magnitude of gravitational potential is 4G.

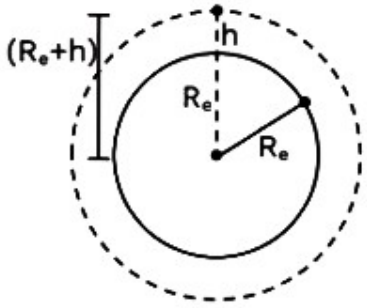
Earth Satellite

A satellite is a body revolving around a planet in circular orbit.

Motion of satellites around a planet is similar to the motion of planets around the Sun. Kepler's laws of planetary motion are equally applicable to the motion of satellites.

Earth's only natural satellite is Moon which takes 27.3 days to complete one circular orbot around Earth.

Apart from the natural satellite, many man-made satellites are revolving Earth for purposes of telecommunication, geo-mapping, astronomy etc.



Consider a satellite having mass m is revolving around Earth at height h above Earth's surface. Let v_0 be the speed of the satellite in its orbit.
According to Newton's law of gravitation, the force of gravitation on a satellite will be

$$F_{(gravitation)} = \frac{GM_em}{(R_e + h)^2} \qquad ...(i)$$

While, the centripetal force required by the satellite to be in orbit is

$$F_{(Centripetal)} = \frac{mv_0^2}{(R_e + h)} \qquad ...(ii)$$

Now, in equilibrium, gravitational pull of the earth provides centripetal force

so, $F_{(gravitation)} = F_{(Centripetal)}$

From equation (i) and (ii), we get

$$\frac{mv_0^2}{(R_e + h)} = \frac{GM_em}{(R_e + h)^2}$$

$$\therefore \quad v_0^2 = \frac{GM}{(R_e + h)}$$

For h = 0,

$$v_0 = \sqrt{\frac{GM_e}{R_e}} = \sqrt{gR_e} \qquad ...(iii) \qquad ...\left(\because \frac{GM_e}{R_e^2} = g\right)$$

This is the minimum speed needed for a satellite to revolve around Earth in an orbit of radius equal to radius of Earth.

Putting the values of g and R_e, we get,

v_0 = 7.92 km/s

Also,

Escape speed, $v_e = \sqrt{2gR_e}$

$\therefore \quad v_e = \sqrt{2}\sqrt{gR_e}$

From equation (iii), we get

$v_e = \sqrt{2}v_0$

So the escape speed is $\sqrt{2}$ times the orbital speed.

If the radius of the orbit of satellite is r, then orbital speed will be given as,

$$v_0 = \sqrt{\frac{GM_e}{r}}$$

Q. The value of escape speed from certain planet is 4 km/s. What will be the orbital speed of satellite orbiting close to its surface?

Sol. As $V_0 = \frac{V_e}{\sqrt{2}}$

$\therefore \quad V_0 = \frac{4}{\sqrt{2}}$

$= 2\sqrt{2}$ km/s

Q. What will be the orbital speed from a planet of mass 10^{30} kg and of radius 10^8 m?

Sol. $V_0 = \sqrt{\frac{GM}{R}}$

$\therefore \quad V_0 = \sqrt{\frac{6.67 \times 10^{-11} \times 10^{30}}{10^8}}$

$\therefore \quad V_0 = \sqrt{66.7 \times 10^{10}}$

$V_0 = 8.1 \times 10^5$ m/s

Energy of an Orbiting Satellite

Kinetic energy of a satellite is, $KE = \frac{1}{2}mv_0^2$

As $v_0 = \sqrt{\frac{GM_e}{(R_e + h)}}$

$\therefore \quad KE = \frac{GM_e m}{2(R_e + h)}$

Also, the potential energy of the satellite will be,

$$PE = -\left(\frac{GM_e m}{r}\right)$$

Total energy = PE + KE = $-\frac{GM_e m}{r} + \frac{GM_e m}{2r} = -\left(\frac{GM_e m}{2r}\right)$

So, the total energy of an orbiting satellite is negative. Since the satellite is bound, so its total energy can never be zero or positive.

Geostationary and Polar Satellites

A geostationary satellite revolves in the equatorial plane of Earth with a time period of 24 hours. So, it appears stationary from a point on Earth.

We can use the relation, $T = \frac{2\pi r}{v_o}$ to find the time period of the satellite.

$$T = \frac{2\pi r}{\sqrt{\frac{GM}{r}}} \Rightarrow T = \frac{2\pi r^{\frac{3}{2}}}{\sqrt{GM}} = \frac{2\pi(R + h)^{\frac{3}{2}}}{\sqrt{GM}}$$

$$\therefore \quad h = \left[\frac{T^2 R_e^2 g}{4\pi^2}\right]^{1/3} - R_e$$

On putting the known values, we get

$$h = 36600 \text{ km (approx)}$$

So, a geostationary satellite must be at a height of 36,600 km above Earth's surface.

Polar satellites revolve at lesser heights (h= 500 – 800 km) above the earth's surface. They have a time period of approximately 100 minutes.

Weightlessness

The force with which Earth attracts a body towards its centre is called weight.

$$W = mg$$

And, $a = g$.

SI unit of weight is Newton.

If a body is in free fall, then the instrument used to measure its weight shows a zero reading. This is called weightlessness.

Q. **Three ball are projected from the surface of Earth with velocities $\sqrt{gR}$, $\sqrt{\frac{3}{2}gR}$, $\sqrt{\frac{5}{4}gR}$, where g = acceleration due to gravity at Earth's surface and R = radius of Earth. Find out the ratio of height achieved by them?**

Sol. $k_1 = \frac{v_1}{v_e} = \frac{1}{\sqrt{2}}$; $k_2 = \frac{v_2}{v_e} = \sqrt{\frac{3}{4}}$; $k_3 = \frac{v_3}{v_e} = \sqrt{\frac{5}{8}}$

From $h = \frac{Rk^2}{1-k^2}$; $h_1 = \frac{R\left(\frac{1}{2}\right)}{1-\frac{1}{2}} = R$, $h_2 = \frac{R\left(\frac{3}{4}\right)}{1-\frac{3}{4}} = 3R$, $h_3 = \frac{R\left(\frac{5}{8}\right)}{1-\frac{5}{8}} = \frac{5R}{3}$

$\therefore \quad h_1 : h_2 : h_3 = 1 : 3 : \frac{5}{3} = 3 : 9 : 5$

Q. **Assuming the radius of the Earth to be 6.4×10^6 m,**

(a) **calculate the time period T of a satellite for equatorial orbit at 1.4×10^6 m above the surface of the Earth.**

(b) **What is the speed of the satellite in this orbit?**

(c) **If the satellite is travelling in the same direction as the rotation of the Earth i.e., West to East, what is the interval between two successive times at which it will appear vertically overhead to an observer at a fixed point on the equator?**

Sol. (a) $r = R_{Earth} + h$

$= 6.4 \times 10^6\ m + 1.4 \times 10^6\ m$

$= 7.8 \times 10^6\ m$

$\Rightarrow T = \left[\frac{4\pi^2 r^3}{GM_{Earth}}\right]^{1/2} = 6831\ s$

(b) Speed of the satellite

$\Rightarrow V = \sqrt{\frac{G.M_{Earth}}{r}} = 7174\ ms^{-1}$

(c) relative angular velocity = $(\omega_{Satellite} - \omega_{Earth})$

Thus, $T = \frac{2\pi}{(\omega_s - \omega_e)}$

$= \frac{2\pi}{\left[\frac{2\pi}{T_s} - \frac{2\pi}{T_e}\right]}$

$\Rightarrow T = \frac{T_s T_e}{T_e - T_s} = 7417\ s$

Q. **If a planet is suddenly stopped in its orbit, supposed to be circular, then show that it would fall into the sun in a time $\frac{T}{4\sqrt{2}}$, where T is the time period of revolution.**

Sol. Using conservation of energy,

$\frac{1}{2}mv^2 - \frac{GMm}{x} = -\frac{GMm}{r}$ (r = radius of the orbit)

$\Rightarrow \frac{1}{2}\left(\frac{dx}{dt}\right)^2 - \frac{GM}{x} = -\frac{GM}{r}$

$\Rightarrow \sqrt{\frac{2GM}{r}}\int_0^t dt = \int_r^0 \sqrt{\frac{x}{r-x}}dx$

$\Rightarrow t = \frac{\pi r}{2}\sqrt{\frac{r}{2GM}} = \frac{T}{4\sqrt{2}}$ $\left[\text{as } T = \left(\frac{4\pi^2 r^3}{GM}\right)^{\frac{1}{2}}\right]$

Q. **Two satellites S_1 and S_2 revolve round a planet in coplanar circular orbits in the same sense. Their periods of revolution are 1 hr and 8 hr respectively. The radius of the orbits of S_1 is 10^4 km. When S_2 is closest to S_1, find**

(a) The speed of S_2 relative to S_1

(b) The angular speed of S_2 as actually observed by an astronaut in S_1

Sol.

(a) $$\frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2} \quad \text{or} \quad \frac{r_2}{r_1} = \left(\frac{T_2}{T_1}\right)^{2/3}$$

$$\Rightarrow \quad \frac{r_2}{r_1} = \left(\frac{8}{1}\right)^{2/3} = 4$$

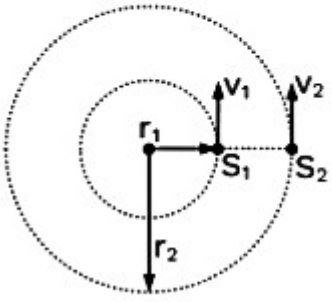
$$\Rightarrow \quad r_2 = 4r_1$$

Given,

$$r_1 = 10^4 \text{ km}$$

$$r_2 = 4 \times 10^4 \text{ km}$$

we get,



$$v_1 = \frac{2\pi r_1}{T_1} = \frac{2\pi \times 10^4}{1} = 2\pi \times 10^4 \text{ km/h}$$

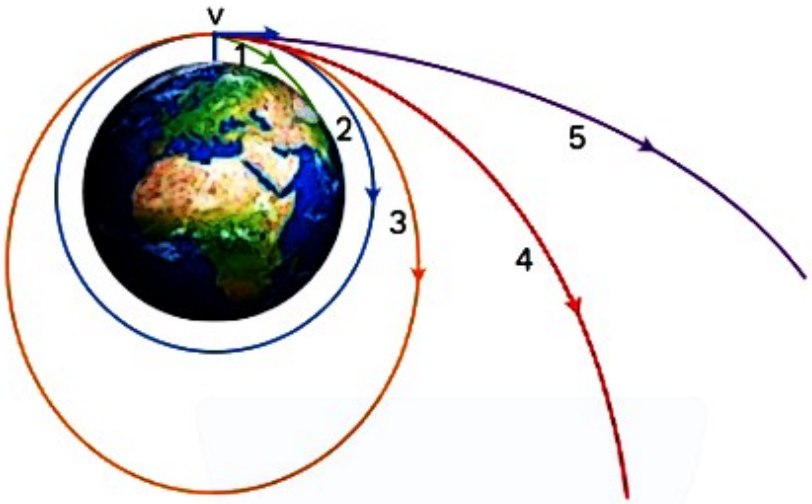
$$v_2 = \frac{2\pi r_2}{T_2} = \frac{2\pi \times 4 \times 10^4}{8} = \pi \times 10^4 \text{ km/h}$$

$\therefore$ Relative speed $v_{2,1} = v_2 - v_1 = -\pi \times 10^4$ km/h

(b) $$\omega_{2,1} = \frac{v_2 - v_1}{r_2 - r_1} = \frac{-\pi \times 10^4 \text{ km/h}}{(4-1) \times 10^4 \text{ km}} = -\frac{\pi}{3} h^{-1}$$

Different projection speeds and trajectories

If a body is projected from Earth's surface with different velocities, then it follows different paths as shown in the figure below.



If the projection speed v is less than $v_o = \sqrt{\frac{GM}{R}}$, then the body follows path 1 and its falls on Earth's surface following an elliptical trajectory.

If the projection speed v is equal to $v_o = \sqrt{\frac{GM}{R}}$, then the body starts revolving around Earth in circular path of radius R shown as 2.

If the projection speed v is such that $\sqrt{\frac{GM}{R}} < v < \sqrt{\frac{2GM}{R}}$, then the body revolves around Earth in elliptical path 3.

If the projection speed v is equal to $\sqrt{\frac{2GM}{R}}$, then the body escapes the gravitational field of Earth following a parabolic trajectory shown as path 4.

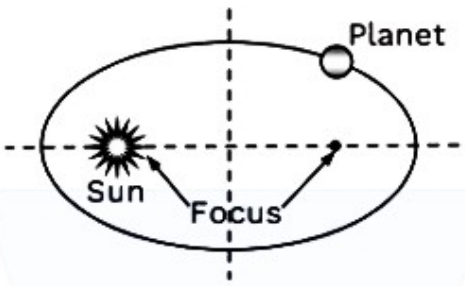
If the projection speed v is more than $\sqrt{\frac{2GM}{R}}$, then the body escapes the gravitational field of Earth following a hyperbolic trajectory shown as path 5.

Kepler's Laws of Planetary Motion (Motion of Planets Around Sun)

Johannes Kepler suggested in his laws that the planets are not revolving around Sun in circular orbits. In fact, the orbits are elliptical and he gave three laws for planetary motion.

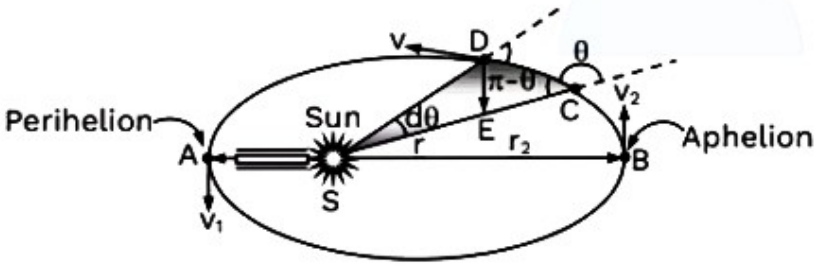
Kepler's First Law (The law of Orbits)

Kepler's first law states that all the planets move around Sun in elliptical orbits with Sun at one of the focus of the ellipse.

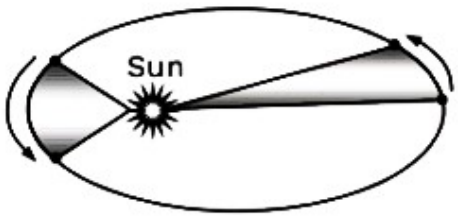


Kepler's Second Law (The law of Areas)

Kepler's second law states that the line which joins the Sun and the planet sweeps out equal area in equal time.



(a)



(b)

$$\frac{dA}{dt} = \text{constant}$$

The second law is just another form of law of conservation of angular momentum.

Consider that the planet is at point C in the orbit and it has a speed v at this instant. Let r be the distance of the planet from Sun and θ be the angle between the velocity vector of planet and its position vector $\vec{r}$.

Then the angular momentum of planet is,

$$L = mvr\sin\theta$$

In small time dt, let the planet reaches the point D.

Then, CD = v.dt

In time dt, position vector $\vec{r}$ sweeps out an area equal to area of the triangle CSD,

Area of triangle CSD = dA = $\frac{1}{2} \times r \times vdt \sin(\pi - \theta) = \frac{1}{2} rv \sin\theta$

So, the rate of sweeping area by the position vector $\vec{r}$ is written as,

$$\frac{dA}{dt} = \frac{1}{2} rv\sin\theta \therefore$$

Also, $rv\sin\theta = \frac{L}{m}$

So, $\frac{dA}{dt} = \frac{L}{2m}$ = constant

Kepler's Third Law (The Law of Periods)

It states that square of time period of planet around Sun is directly proportional to the cube of the semi-major axis of the elliptical path around Sun.

If T = time period of revolution, and a = semi-major axis, then

$$T^2 \propto a^3$$

For circular orbits, major and minor axis are equal.

If a planet is in a circular orbit of radius r , then its revolution speed is

$$v = \sqrt{\frac{GM_s}{r}}$$

where M_s is the mass of sun.

Then, $T = \frac{2\pi r}{v}$

or $T = \frac{2\pi r}{\sqrt{\frac{GM_s}{r}}}$

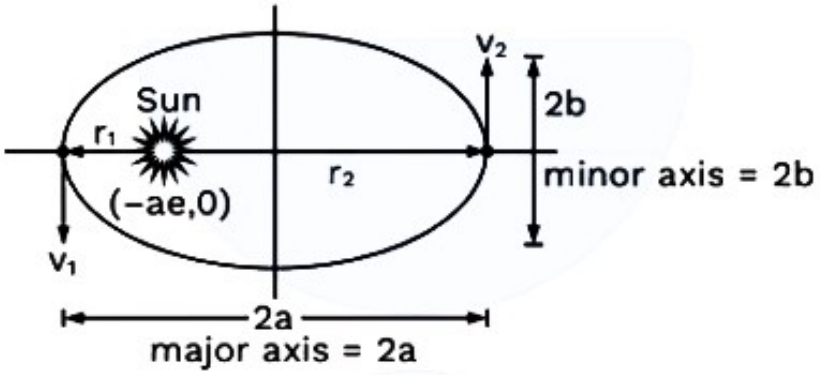
On squaring the above equation,

$$T^2 = \frac{4\pi^2}{GM_s} r^3 \Rightarrow T^2 \propto r^3$$

For elliptical orbits, we can write,

$$\frac{dA}{dt} = \frac{L}{2m}$$

Let us consider the elliptical path of planet shown in figure with sun at one focus (−ae, 0).



Here r_1 and r_2 are the shortest and farthest distance of planet from sun during its motion, which are given as

$$r_1 = a(1 - e)$$

and $r_2 = a(1 + e)$

where e is the eccentricity.

From geometry of ellipse,

$$b = a\sqrt{1 - e^2}$$

If v_1 = Speed at perihelion, and v_2 = Speed at aphelion, then by conservation of angular momentum,

$$L = mv_1r_1 = mv_2r_2$$

By energy conservation,

$$\frac{1}{2}mv_1^2 - \frac{GM_sm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GM_sm}{r_2}$$

or $v_1^2 - v_2^2 = 2GM_s\left[\frac{1}{r_1} - \frac{1}{r_2}\right]$

$$v_1^2\left[1 - \frac{r_1^2}{r_2^2}\right] = 2GM_s\left[\frac{r_2 - r_1}{r_1 r_2}\right]$$

or $v_1 = \sqrt{\frac{2GM_s r_2}{(r_1 + r_2)r_1}}$, then $v_2 = \sqrt{\frac{2GM_s r_1}{(r_1 + r_2)r_2}}$

Also, by Kepler's second law, we can write

$$A = \frac{L}{2m}T$$

or $T = \frac{2m}{L}A = \frac{2m(\pi ab)}{L} = \frac{2m(\pi ab)}{mv_1 r_1}$

or $T = \frac{2m\pi a\left[a\sqrt{1-e^2}\right]}{m\left[\sqrt{\frac{GM_s}{a}\left(\frac{1+e}{1-e}\right)}\right][a(1-e)]}$

or $T^2 = \frac{4\pi^2}{GM_s}a^3 \Rightarrow T^2 \propto a^3$

Q. **The semi-major axis of the orbit of Saturn is approximately nine times that of Earth. The time period of revolution of Saturn is approximately equal to?**

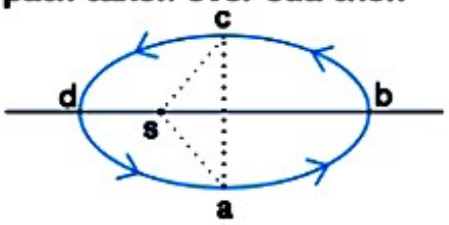
Sol. $R_S = 9R_E$

Using law of periods,

$$\left(\frac{T_S}{T_E}\right)^2 = \left(\frac{R_S}{R_E}\right)^3$$

$$T_S = T_E\left(\frac{R_S}{R_E}\right)^{\frac{3}{2}} = T_E\,(9)^{\frac{3}{2}} = 27\,T_E = 27 \text{ years}$$

Q. **Figure shows elliptical path abcd of a planet around the sun S such that the area of triangle csa is $\frac{1}{4}$ the area of the ellipse. With db as the semimajor axis, and ca as the semiminor axis, if t_1 is the time taken for planet to go over path abc and t_2 for path taken over cda then**



(A) $t_1 = 4t_2$ **(B)** $t_1 = 2t_2$ **(C)** $t_1 = 3t_2$ **(D)** $t_1 = t_2$

Sol. Let the area of the ellipse be A.

As per Kepler's 2nd law, areal velocity of planet aorund the sun in constat,

$$\text{i.e., } \frac{dA}{dt} = \text{constant.}$$

$$\therefore \frac{t_1}{t_2} = \frac{\text{Area of absca}}{\text{Area of adcsa}}$$

$$\frac{\frac{A}{2}+\frac{A}{4}}{\frac{A}{2}-\frac{A}{4}} = \frac{\frac{3A}{4}}{\frac{A}{4}} = 3$$

$$\Rightarrow t_1 = 3t_2$$

Note: Here db is the major axis of the ellipse, not semimajor axis and ca is the minor axis of the ellipse, not semiminor axis.

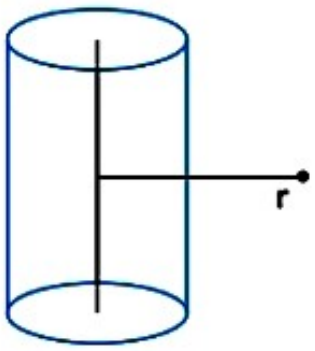
Q. **A very long (length L) cylinderical galaxy is made of uniformly distributed mass and has radius R(R < < L). A star outside the galaxy is orbiting the galaxy in a plane perpendiucalr to the galaxy and passing through its centre. If the time period of star is T and its distance from the galaxy's axis is r, then**

(A) $T^2 \propto r^3$ **(B)** $T \propto r^2$ **(C)** $T \propto r$ **(D)** $T \propto \sqrt{r}$

Sol. Centripetal force is proveded by gravitational force

$$F = \frac{2GM}{Lr}m \Rightarrow \quad F = \left(\frac{k}{r}\right)m$$

Here, k = constant.



So, $\frac{mv^2}{r} = \frac{km}{r}$

$\Rightarrow v =$ constant

$T = \frac{2\pi r}{v}$.

$\therefore T \propto r$

Q. **Average distance of the earth from the sun is L_1. If one year of the earth = D days, one years of another planet whose average distance from the sun is L_2 will be**

(A) $D\left(\frac{L_2}{L_1}\right)^{1/2}$ days

(B) $D\left(\frac{L_2}{L_1}\right)^{3/2}$ days

(C) $D\left(\frac{L_2}{L_1}\right)^{2/3}$ days

(D) $D\left(\frac{L_2}{L_1}\right)$ days

Sol. According to kepler's third law,

$T^2 \propto r^3$ or $T \propto r^{3/2}$

$\therefore \frac{D'}{D} = \left(\frac{L_2}{L_1}\right)^{3/2}$ or $D' = D\left(\frac{L_2}{L_1}\right)^{3/2}$ days

Q. **Two particles of masses m and 2m are placed at a distance of r apart. Initially 2m is given a velocity v_0 away from m and m is at rest (but free to move). Find the maximum separation between the particles.**

Sol. At the instance of maximum separation both must move with equal speed v (say).

Applying linear momentum conservation

$$2mv_0 = (2m + m)v \qquad \Rightarrow \qquad v = \frac{2}{3}v_0$$

By energy conservation (if maximum separation is r'), we get

$$-\frac{G2m^2}{r} + \frac{1}{2}2mv_0^2 = -\frac{G2m^2}{r'} + \frac{1}{2}(2m+m)\left(\frac{2}{3}v_0\right)^2$$

$$\Rightarrow \quad -\frac{G2m^2}{r} + mv_0^2 - \frac{2}{3}mv_0^2 = -\frac{G2m^2}{r'}$$

$$\Rightarrow \quad -\frac{G2m}{r} + \frac{1}{3}v_0^2 = -\frac{G2m}{r'}$$

$$\Rightarrow \quad \frac{2Gm}{r'} = \frac{2Gm}{r} - \frac{v_0^2}{3}$$

$$\Rightarrow \quad r' = \frac{2Gm}{\left(\frac{2Gm}{r} - \frac{v_0^2}{3}\right)}$$

Q. **The distance of two planets from the sun are 10^{13} m and 10^{12} m respectively. What is the ratio of their time periods?**

Sol. As $T^2 \propto R^3$ ...(By Kepler's third Law)

$$\therefore \quad \frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3}$$

$$\therefore \quad \frac{T_1^2}{T_2^2} = \left[\frac{10^{13}}{10^{12}}\right]^3$$

$$\therefore \quad \left[\frac{T_1}{T_2}\right]^2 = 10^3$$

$$\therefore \quad \frac{T_1}{T_2} = \frac{10\sqrt{10}}{1}$$

Q. **A geostationary satellite is orbiting the earth at a height 6R above the surface of earth, where R is radius of earth. What will be the time period of another satellite at a height 2.5 R from the surface of earth?**

Sol. As $T^2 \propto R^3$ ...(By Kepler's third Law)

$$\therefore \quad \frac{T_1^2}{T_2^2} = \frac{R_1^3}{R_2^3}$$

$$\text{or} \quad \left(\frac{T_2}{T_1}\right)^2 = \left(\frac{R_2}{R_1}\right)^3$$

$$\Rightarrow \quad T_2^2 = T_1^2\left[\frac{3.5R}{7R}\right]^3$$

$$\Rightarrow \quad T_2^2 = \frac{T_1^2}{8}$$

$$\Rightarrow \quad T_2 = \frac{T_1}{2\sqrt{2}}$$

$$T_2 = \frac{24}{2\sqrt{2}} \quad \text{...}(\because T_1 = 24\,\text{hours})$$

$$\therefore \quad T_2 = 6\sqrt{2}\,\text{hours}$$